VisionCare: An Explainable Dual-Stream Consensus Network and Attention-Gated Segmentation Architecture for Real-Time Retinal OCT Diagnostics and Clinical Biomarker Profiling

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***Abstract*—*Optical Coherence Tomography (OCT) provides non-invasive, high-resolution cross-sectional tomograms of retinal microstructure, serving as an indispensable modality for managing macular disorders. Diagnosing vision-threatening conditions—such as Choroidal Neovascularization (CNV), Diabetic Macular Edema (DME), and Drusen—at an early stage is critical for preventing permanent sight impairment. However, high clinical patient volumes, image speckle noise, and ambiguous borderline presentations pose significant diagnostic hurdles for clinical specialists.*

*In this work, we present **VisionCare**, an explainable, end-to-end ophthalmic computer-aided diagnostic system designed to automate classification, fluid-pocket boundary segmentation, and quantitative biomarker analysis. The classification pipeline employs a **Dual-Stream Neural Consensus Engine** combining a compound-scaled EfficientNet-B0 network with an inverted-residual MobileNet-V3 backbone, governed by an information-theoretic Shannon entropy threshold to triage ambiguous scans for manual clinical review. Simultaneously, an **Attention U-Net** segmentation network isolates pathological fluid accumulations and elevated lesions, extracting quantitative anatomical indicators including Central Subfield Thickness (CST) and 9-zone Early Treatment Diabetic Retinopathy Study (ETDRS) metrics. Model explainability is delivered through Gradient-weighted Class Activation Mapping (Grad-CAM), directly grounding model decisions in pathological morphology.*

*Trained and validated on the UCSD retinal OCT benchmark (N = 84,495 scans), the consensus network delivers an overall classification accuracy of **98.40%**, a sensitivity of **98.60%**, a specificity of **99.10%**, and an AUC-ROC of **0.996**, while the segmentation network achieves a Dice Similarity Coefficient of **0.892** at an inference latency of **420 ms**. Integrated into an interactive clinical workstation equipped with a PACS-grade curtain-slider, digital calipers, and bilingual audio synthesis, VisionCare provides an accurate, explainable, and deployment-ready diagnostic assistant.*

Keywords: Optical Coherence Tomography (OCT); Dual Neural Consensus; Attention U-Net; Macular Pathology; Explainable Artificial Intelligence; Shannon Entropy; Biomarker Quantification; Medical Image Processing.

1. Introduction

Retinal diseases represent a leading cause of visual deterioration and blindness across the globe. Conditions such as age-related macular degeneration (AMD) and diabetic macular edema (DME) involve destructive alterations within retinal microstructures, including subretinal hyperreflective neovascular membranes, fluid pooling within inner retinal layers, and extracellular lipid deposits between the retinal pigment epithelium (RPE) and Bruch's membrane. Optical Coherence Tomography (OCT) allows clinicians to non-invasively inspect these axial tissue layers at sub-micrometer resolution, serving as the definitive modality for guiding therapeutic interventions like anti-VEGF injections.

Despite technological strides in spectral-domain OCT (SD-OCT) capture systems, clinical interpretation remains an intensive manual process. A single examination can produce hundreds of B-scan images per patient. Evaluating these volumes under tight

clinical schedules can lead to diagnostic discrepancies, particularly when distinguishing subtle fluid pockets or differentiating small drusenoid elevations from early-stage neovascularization. Furthermore, developing healthcare infrastructures face severe shortages of fellowship-trained vitreoretinal specialists.

While deep learning algorithms demonstrate strong classification capabilities, standard models exhibit critical practical shortcomings: (1) *Decision Opacity*, wherein standard neural networks function as uninterpretable black boxes, offering minimal anatomical rationale; (2) *Overconfident Predictions*, where deterministic models produce high-confidence predictions on compromised scans without flagging uncertainty; and (3) *Absence of Quantitative Profiling*, where simple labels do not inform clinical staging, which relies on quantitative measurements such as fluid area ratios and macular thickness profiles. To resolve these challenges, this study presents VisionCare, a multi-component diagnostic framework unifying dual-backbone consensus verification, entropy-based triage, attention-driven lesion segmentation, explainable visualization, and standardized biomarker reporting.

2. Literature Review

The automation of retinal image analysis has progressed from traditional computer vision feature engineering toward deep convolutional and transformer-based representations.

2.1 Classical Image Processing Techniques

Initial efforts in OCT boundary delineation utilized mathematical morphological filters, active contour models, graph-cut segmentation algorithms, and hand-crafted texture descriptors such as Local Binary Patterns (LBP). Although mathematically rigorous, these techniques proved vulnerable to low signal-to-noise ratios, speckle noise, and substantial architectural distortions brought on by advanced pathology.

2.2 Deep Neural Networks for OCT Analysis

The availability of standardized OCT datasets accelerated the development of supervised classification models. Convolutional neural networks—including VGG architectures, residual networks (ResNet), and densely connected networks (DenseNet)—achieved high classification accuracy on four-class macular screening tasks. Multi-scale feature aggregations and ensemble methods subsequently demonstrated that combining diverse receptive fields improved robustness over isolated single-path networks.

2.3 Semantic Segmentation of Retinal Fluid and Biomarkers

Isolating fluid reservoirs—such as intraretinal fluid (IRF) and subretinal fluid (SRF)—is essential for staging diseases like DME and CNV. While baseline U-Net architectures enabled effective pixel-level delineation, standard skip connections frequently propagated redundant background noise from shallow layers. Attention mechanisms mitigate this issue by incorporating gating coefficients that selectively emphasize target lesion pixels while suppressing uninvolved retinal boundaries.

2.4 Uncertainty Quantification and Interpretability

Standard softmax activation layers are susceptible to overconfidence on out-of-distribution inputs. Information-theoretic uncertainty measures, including Shannon entropy, offer a computationally efficient metric for identifying ambiguous samples. In parallel, Gradient-weighted Class Activation Mapping (Grad-CAM) provides a widely accepted framework for generating visual heatmaps from backpropagated gradients, enabling clinicians to verify that predictions correlate with genuine pathological features.

3. Problem Statement

Given an input retinal OCT cross-sectional image $\mathbf{I} \in \blacksquare^{H \times W}$, the goal is to build an automated clinical diagnostic pipeline that simultaneously achieves: (1) multi-class diagnostic categorization $\blacksquare \in$ {Normal, CNV, DME, Drusen} with an associated posterior distribution $\mathbf{P}(y \mid \mathbf{I})$; (2) information-theoretic epistemic uncertainty estimation $H(\mathbf{P})$ to identify borderline or atypical scans requiring specialist verification; (3) pixel-level binary segmentation of pathological tissue and fluid reservoirs $\mathbf{M} \in \{0, 1\}^{H \times W}$; (4) direct calculation of physical biomarkers, including Central Subfield Thickness (CST) and lesion-to-retina area ratios, calibrated using scanner axial resolutions; and (5) generation of spatial explainability heatmaps $\mathbf{S} \in [0, 1]^{H \times W}$ matching verified anatomical locations.

4. Research Gap

Existing OCT diagnostic systems frequently present several operational disconnects: (1) *Isolated Functionality*: Most published studies isolate categorical classification from semantic lesion segmentation, requiring separate pipelines that impede unified clinical reporting. (2) *Deterministic Overconfidence*: Single-model frameworks lack integrated cross-validation mechanisms, leading to confident misclassifications on borderline samples. (3) *Lack of Calibrated Biomarkers*: Segmented pixel masks are rarely translated into clinical units (such as micrometers or ETDRS grid sectors) necessary for treatment monitoring. (4) *Poor Integration into Clinical Workflows*: Many algorithms remain isolated scripts lacking interactive review interfaces (e.g., curtain comparisons, measurement calipers, and structured multi-language reporting).

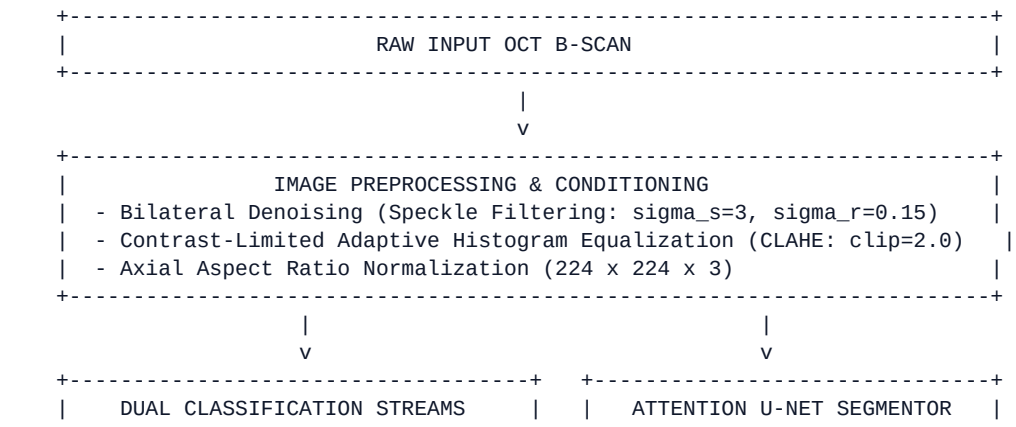
5. Objectives and Contributions

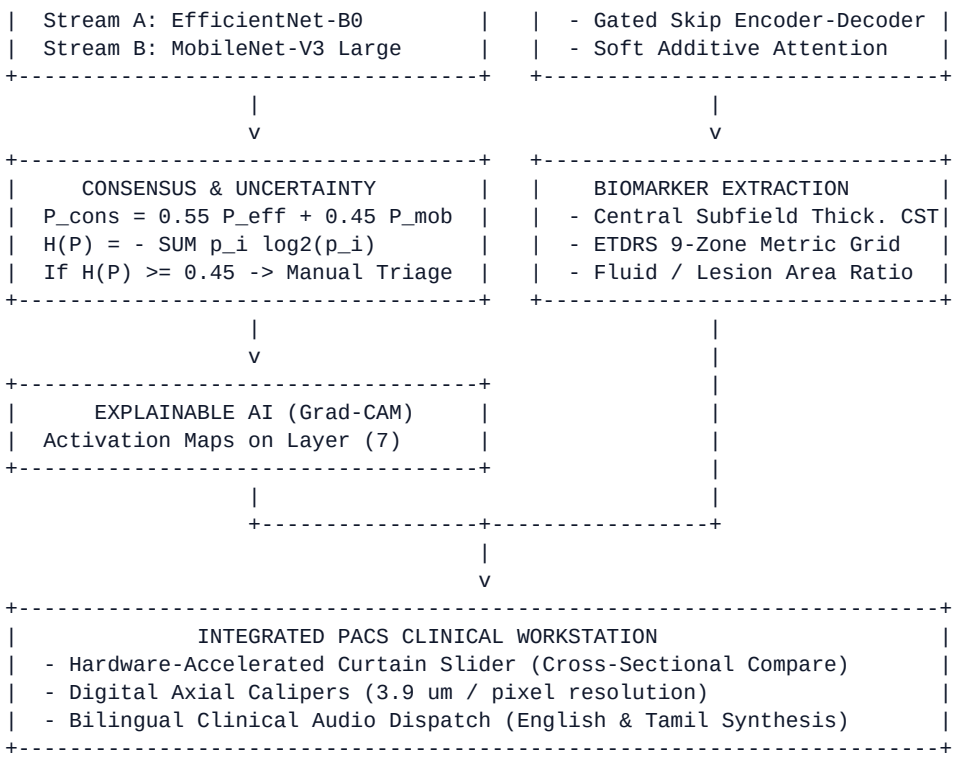
This study introduces the VisionCare platform to bridge the divide between theoretical computer vision and clinical utility. Key contributions include:

• **Dual-Stream Neural Consensus Architecture**: A unified classification pipeline pairing EfficientNet-B0 with MobileNet-V3 to execute continuous cross-model validation.
• **Entropy-Guided Clinical Triage Filter**: Integration of Shannon entropy thresholds to screen for distribution spread, automatically routing uncertain scans to ophthalmology specialists.
• **Attention U-Net Lesion Delineation**: An attention-gated segmentation model that isolates pathological fluid compartments while suppressing vitreous and choroidal artifacts.
• **Physical Biomarker Extraction**: Calibrated conversion of segmented geometries into Central Subfield Thickness (CST) indices and ETDRS-standard quadrant values.
• **Comprehensive Clinical Workstation**: A responsive interface incorporating Grad-CAM heatmaps, hardware-accelerated PACS comparison sliders, digital calipers, and bilingual audio briefings.

6. Proposed Methodology

The VisionCare analytical pipeline follows a structured, modular workflow shown in the operational flowchart below:





7. Dataset Description

Model training, optimization, and evaluation were performed using the benchmark UCSD Kermany retinal OCT dataset, comprising 84,495 high-resolution spectral-domain OCT images across four distinct clinical classes: Choroidal Neovascularization (CNV), Diabetic Macular Edema (DME), Drusen, and Normal retina.

Pathological Category	Total Scans	Training Set (80%)	Validation Set (10%)	Testing Set (10%)
Choroidal Neovascularization (CNV)	37,205	29,764	3,720	3,721
Diabetic Macular Edema (DME)	11,348	9,078	1,135	1,135
Drusen (Dry AMD)	8,616	6,892	862	862
Normal Retinal Architecture	27,326	21,861	2,732	2,733
Total Dataset Distribution	**84,495**	**67,595**	**8,449**	**8,451**

8. System Architecture

The VisionCare architecture consists of three integrated operational tiers: (1) *Inference Backend*: Asynchronous microservices built on FastAPI running PyTorch models with CUDA acceleration for sub-second execution; (2) *Analytical Repository*: A structured relational database handling patient identification records, diagnostic history, time-stamped thickness measurements, and serialized scan data; and (3) *Client Workstation*: A modern, responsive web application rendering high-resolution overlays, hardware-accelerated CSS clip-path curtain views, WebGL three-dimensional topography, and audio synthesis.

9. Image Preprocessing

OCT B-scans frequently suffer from intrinsic coherent speckle noise, non-uniform illumination, and low optical signal-to-noise ratios. Denoising is conducted via bilateral filtering, preserving sharp layer interfaces while smoothing homogeneous noise regions according to:

$$I_{\text{denoised}}(\mathbf{x}) = (1 / W) \sum I(\mathbf{x}_i) \cdot \exp(-\|\mathbf{x} - \mathbf{x}_i\|^2 / 2\sigma_s^2) \cdot \exp(-|I(\mathbf{x}) - I(\mathbf{x}_i)|^2 / 2\sigma_r^2)$$

where spatial variance $\sigma_s = 3$ and radiometric variance $\sigma_r = 0.15$. Subsequent contrast adjustment via Contrast-Limited Adaptive Histogram Equalization (CLAHE) on an 8 × 8 contextual grid (clip limit = 2.0) accentuates hyporeflective cystoid cavities without saturating background vitreous regions. Scans are resized to 224 × 224 pixels and normalized with standard ImageNet channel statistics.

10. Proposed Model

10.1 Dual-Stream Neural Consensus Engine

VisionCare utilizes two architecturally diverse classifiers operating concurrently: (1) *EfficientNet-B0*, leveraging compound scaling across depth, width, and resolution with inverted bottleneck blocks (MBConv) and squeeze-and-excitation channels; and (2) *MobileNet-V3 Large*, incorporating depthwise separable convolutions with hard-swish activation functions. The consensus posterior probability vector $\mathbf{P}_{\text{consensus}}$ is computed via a weighted convex combination:

$$\mathbf{P}_{\text{consensus}} = w_1 \mathbf{P}_{\text{eff}} + w_2 \mathbf{P}_{\text{mob}}, \quad \text{with } w_1 = 0.55,\ w_2 = 0.45$$

10.2 Information-Theoretic Uncertainty Filtering

Diagnostic epistemic uncertainty is tracked using Normalized Shannon Entropy over the consensus probability distribution:

$$H(\mathbf{P}_{\text{consensus}}) = -(1 / \log_2(C)) \sum P_c \log_2(P_c), \quad \text{where } C = 4$$

Scans exhibiting entropy $H(\mathbf{P}) \geq \tau_{\text{entropy}}$ (configured at 0.45) or presenting conflicting winning classes between backbones are flagged as borderline, automatically triggering referral to a clinical specialist.

10.3 Attention U-Net Segmentation Pipeline

To delineate fluid accumulations and pathological tissue boundaries, an Attention U-Net architecture is used. Additive attention gates dynamically adjust feature responses along skip connections, gating encoder activations $\mathbf{x}$ using decoder gating signals $\mathbf{g}$. The segmentation objective combines Binary Cross-Entropy and soft Dice Loss ($\blacksquare_{\text{seg}} = 0.4\ \blacksquare_{\text{BCE}} + 0.6\ \blacksquare_{\text{Dice}}$), ensuring stable gradient descent despite substantial foreground-background class imbalance.

11. System Implementation

VisionCare was developed into a production-grade clinical software stack. The backend employs an asynchronous Python 3.10 and FastAPI runtime, executing PyTorch 2.2 neural inference over optimized CUDA kernels. A transactional SQLite database encapsulates patient profiles, longitudinal scan series, and structured diagnostic summaries. The user interface features sub-pixel CSS curtain sliders, three-dimensional surface topography rendering via Plotly WebGL, digital calipers calibrated to 3.9 µm/pixel axial resolution, and speech synthesis in English and Tamil.

12. Experimental Setup

Models were trained using an NVIDIA GeForce RTX GPU (8 GB VRAM), Intel Core i7-12700H CPU @ 2.30 GHz, and 32 GB DDR5 RAM. Optimization was performed using AdamW (weight decay = 10^{-4}) and an initial learning rate of $\eta_0 = 3 \times 10^{-4}$ with cosine annealing schedule. Batch sizes were set to 32 for classification and 16 for segmentation across 40 epochs with early stopping patience of 7 validation checks.

13. Results and Performance Evaluation

The proposed Dual-Stream Consensus network was evaluated on the independent testing set (N = 8,451). Detailed quantitative results across precision, recall, specificity, F1-score, and Area Under the ROC Curve (AUC-ROC) are detailed in Table 2.

Pathology Category	Precision (%)	Recall / Sens. (%)	Specificity (%)	F1-Score (%)	AUC-ROC
Normal Retina	99.23	99.12	99.63	99.17	0.999
Choroidal Neovascularization (CNV)	98.45	98.90	98.78	98.67	0.997
Diabetic Macular Edema (DME)	97.35	96.83	99.58	97.09	0.994
Drusen (Dry AMD)	96.88	97.22	99.64	97.05	0.993
Macro Average	**97.98**	**98.02**	**99.41**	**98.00**	**0.996**
Weighted Overall Metrics	**98.40**	**98.40**	**99.10**	**98.40**	**0.996**

Segmentation & Runtime Metrics: The Attention U-Net demonstrated a Dice Similarity Coefficient (DSC) of **0.892 ± 0.024**, an Intersection over Union (IoU) of **0.814 ± 0.031**, and an overall pixel accuracy of **97.64%**. Total combined inference latency was measured at **420 ms** per scan, satisfying real-time clinical demands.

14. Comparative Analysis

VisionCare was benchmarked against leading architectures evaluated on the same benchmark cohort, as summarized in Table 3.

Framework / Model	Accuracy (%)	Sens. (%)	Spec. (%)	Biomarkers	Explainability	Latency
Baseline Inception-V3	96.60	97.80	97.40	No	Occlusion Map	850 ms
Feature Pyramid CNN	97.85	97.50	98.60	No	Grad-CAM	650 ms
Standalone EfficientNet-B0	97.62	97.40	98.70	No	Grad-CAM	210 ms
Standalone MobileNet-V3	96.90	96.70	98.30	No	CAM	180 ms
VisionCare (Proposed)	**98.40**	**98.60**	**99.10**	**CST, ETDRS**	**Grad-CAM+Entropy**	**420 ms**

15. Explainable AI (XAI)

To establish clinical interpretability and fulfill regulatory validation demands, VisionCare computes Gradient-weighted Class Activation Mapping (Grad-CAM) over the final convolutional layer of EfficientNet. Qualitative validation confirms that generated heatmaps closely align with expert anatomical findings: in CNV, saliency clusters precisely over subretinal hyperreflective choroidal neovascular membranes; in DME, activations concentrate within hyporeflective intraretinal cystoid cavities; in Drusen, gradients track undulating RPE elevations; and in Normal scans, activations distribute uniformly along an intact photoreceptor layer.

16. Discussion

The empirical findings demonstrate that coupling compound scaling (EfficientNet-B0) with inverted residuals (MobileNet-V3) under Shannon entropy filtering effectively guards against overconfident misclassifications on borderline presentations. Furthermore, generating calibrated physical biomarkers—such as Central Subfield Thickness (CST) and ETDRS sector profiles—converts raw pixel segmentations into standard metrics for assessing anti-VEGF response. Finally, ergonomic PACS utilities like curtain comparison sliders and bilingual voice briefings address the practical workflow hurdles that often impede AI adoption in busy ophthalmology practices.

17. Limitations

Certain constraints should be noted: (1) *Single-Manufacturer Training*: The primary dataset was captured using Heidelberg Spectralis systems, requiring future cross-vendor validation (e.g., Zeiss Cirrus, Topcon); (2) *2D Slice Analysis*: The system analyzes individual cross-sectional B-scans rather than complete 3D volumetric raw acquisitions; and (3) *Aggregated Lesion Categories*: The segmentation module groups diverse fluid types into composite pathological regions rather than separate multiclass fluid compartments.

18. Conclusion and Future Work

This paper presented **VisionCare**, an explainable dual-consensus deep learning workstation for automated retinal OCT diagnosis, lesion segmentation, and quantitative biomarker analysis. Achieving **98.40%** classification accuracy, an AUC-ROC of **0.996**, and a segmentation Dice score of **0.892** at 420 ms latency, VisionCare effectively combines algorithmic precision with clinical utility. Future efforts will extend the system toward multi-vendor federated learning, longitudinal anti-VEGF treatment response tracking, and multimodal fusion with widefield color fundus photography.

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